

32548 Cyber Security - Week 4 Lab: Cryptography, SSL & X.509

32548 Cyber Security – Week 4 Lab: Cryptography, SSL & X.509 Certificates

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Tutorial: Tut3 | **Subject:** 32548 | **Group:** Cmp1 (Activity 03)

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Task 1 – Symmetric Encryption & Decryption

Symmetric encryption basically mean you use one single shared key for both locking and unlocking the data. For this one i used AES-128 with the CBC mode. The main thing that i noticed is that after encryption the file become totally unreadable, and to get the original back you need the exact same key and IV, if not it just wont work.

Q1 – Text file (Text_file.txt)

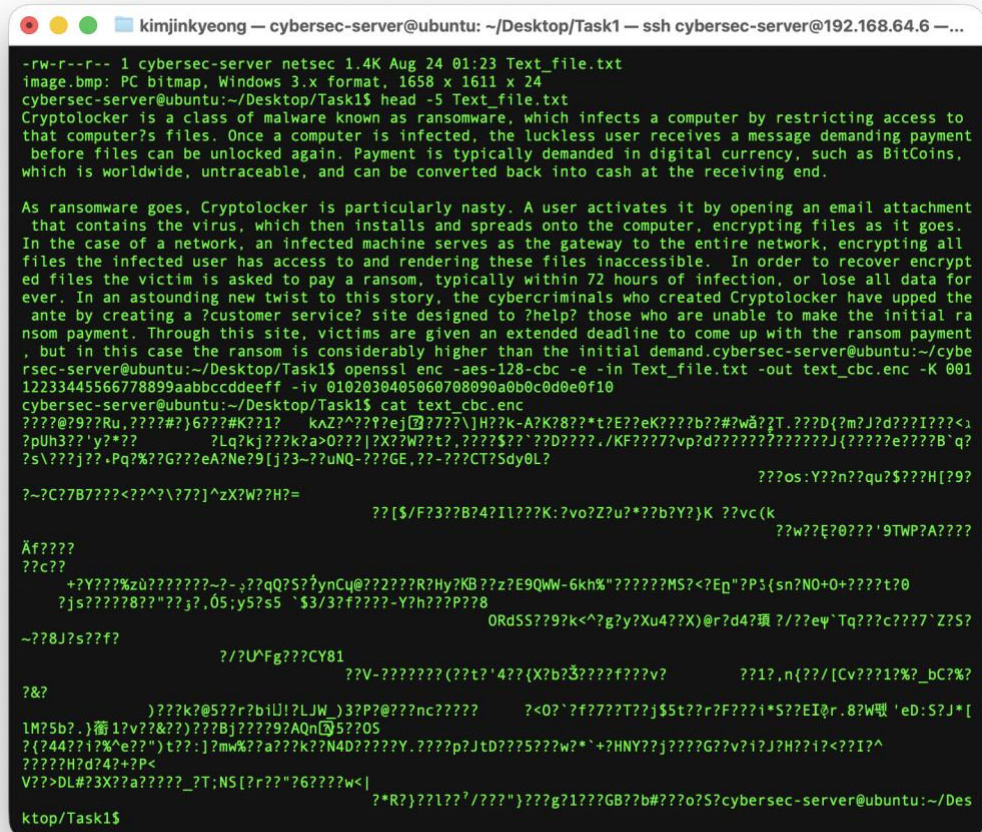
Commands i used for the encryption and also the decryption:

```
openssl enc -aes-128-cbc -e -in Text_file.txt -out text_cbc.enc -K
```

```
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10 openssl
```

```
enc -aes-128-cbc -d -in text_cbc.enc -out text_dec.txt -K
```

```
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/Task1 — ssh cybersec-server@192.168.64.6 —...
-rw-r--r-- 1 cybersec-server netsec 1.4K Aug 24 01:23 Text_file.txt
image.bmp: PC bitmap, Windows 3.x format, 1658 x 1611 x 24
cybersec-server@ubuntu:~/Desktop/Task1$ head -5 Text_file.txt
Cryptolocker is a class of malware known as ransomware, which infects a computer by restricting access to
that computer's files. Once a computer is infected, the luckless user receives a message demanding payment
before files can be unlocked again. Payment is typically demanded in digital currency, such as BitCoins,
which is worldwide, untraceable, and can be converted back into cash at the receiving end.

As ransomware goes, Cryptolocker is particularly nasty. A user activates it by opening an email attachment
that contains the virus, which then installs and spreads onto the computer, encrypting files as it goes.
In the case of a network, an infected machine serves as the gateway to the entire network, encrypting all
files the infected user has access to and rendering these files inaccessible. In order to recover encrypt
ed files the victim is asked to pay a ransom, typically within 72 hours of infection, or lose all data for
ever. In an astounding new twist to this story, the cybercriminals who created Cryptolocker have upped the
ante by creating a ?customer service? site designed to ?help? those who are unable to make the initial ra
nsom payment. Through this site, victims are given an extended deadline to come up with the ransom payment
, but in this case the ransom is considerably higher than the initial demand.cybersec-server@ubuntu:~/cybe
rsec-server@ubuntu:~/Desktop/Task1$ openssl enc -aes-128-cbc -e -in Text_file.txt -out text_cbc.enc -K 001
12233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ cat text_cbc.enc
????@9??Ru,?????#?6??#K??1? kAZ?^??y?e?[]???\H??K-A?K?8??*t?E??eK????b??#?wã??T.???D{?m?J?d???I???<1
?pUh3??'y'?*?? ?Lq?k?j??k?>a?0??|?X??W??t?,???$?? ?D?D???,/KF??vp?d????????????J{????e????B`q?
?s\???j??.Pq%??G??eA?Ne?9[j?3--?uNq-???GE,??-???CT?Sdy0L?

???os:Y??n??qu$??H[?9?
?~?C?7B7??<??^?\?7?]^zX?W?H?=

??[?$/F?3??B?4?I?L??K:??vo?Z?u?*??b?Y?Y?K ??vc(k
??w??E?0???'9TWP?A????
Åf????
??c??
+?Y???%z0??????-?-?~?qQ?S?7ynCu@??????R?Hy?KB??z?E9QWw-6kh%"?????MS?<?Eh"?P3{sn?N0+0+????t?0
?js????8??"??j?,05:y5?s5 `3/3?f????-Y?h????P??8
ORD5S?9?k<^?g?y?Xu4?X?)@r?d4?瓊 ?/?eψ`Tq?c?????`Z?S?
~??8J?s??f? ?/?U^Fg??CY81
??V-???????(??t?'4??{X?b?3?7??f??v? ??1?,n{?/[Cv???1?%?_bC?%?
?&?
)???k?@5??r?b!j!LJW_)3?P?@???nc????? ?<0?'?f????T??j$5t??r?F???i*S???EI?r.8?W?e`eD:S?J*[
lM?5b?,}衛1?v?&??)??Bj????9?AQn[5?7OS
?{44?7i?%"e??")t??:]mW%?7a??k?N4D?????Y.????p?JtD??75?7?w?*`+?HNY??j????G??v?1?J?H?7i?<?I?^
?????H?d?4?+P<
V??>DL#?3X?7a?????_T;NS[?r???"6?????w<|
?R?}?7l??'/???"}??g?1??7GB??b#??o?S?cybersec-server@ubuntu:~/Des
ktop/Task1$
```

Fig 1. The original text, then the encrypt command, and cat text_cbc.enc showing the scrambled ciphertext - now you cant read the text anymore.

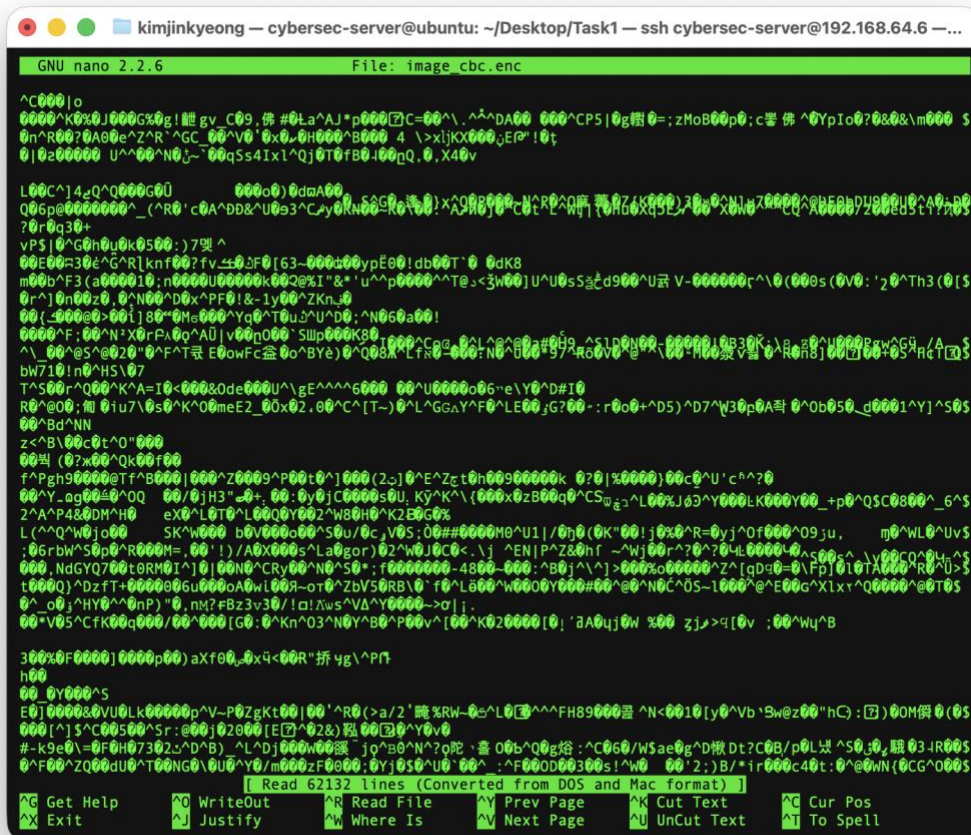
Q2 – Image file (image.bmp)

Same cipher but this time on the image file. Because the encrypted one is a binary i opened it with nano instead (if you open a binary with cat the terminal just get all messed up):

```
openssl enc -aes-128-cbc -e -in image.bmp -out image_cbc.enc -K
```

```
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10 openssl  
enc -aes-128-cbc -d -in image_cbc.enc -out image_dec.bmp -K
```

```
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



```
GNU nano 2.2.6 File: image_cbc.enc

^C00|o
0000^K0%0J000G%0! 卐 gv_C09,佛 #0La^AJ^p000C=00^\\,^A^DA00 000^CP5|0g甯0=:zMoB00p0:c甯佛 ^0YpIo0?0&0&\m000 $
0n^R00?0A00e^Z^R^GC 00^V0'0x0_0H000^B000 4 \>x|jKX000jER!0t;
0|0z00000 U^00^N0j~^00q5s4Ix1^0qj0T0fB0:00nQ,0_X40v

L00C^]4z0^Q000G00 000o0)0dmA00
Q06p00000000^_(^R0'c0A^0D&^U003^C^y0Rn00=Ro^00)^A0B0000^U^R000 卐0ZK000)300^X0W0Z0000^00E0bDU00U00^A00^00
?0r0q30+
vP5|0^c0n0u0k0500:)7甯^
00E000300^G^R|knf00?fv.卐0JF0[63-000000ypE00|db00T 0 dK8
m00b^F3(a000010:n0000U00000k00200I"&+^u^p0000^ATe.<3w00)U^U0s50c^d900^U00 V-000000r^0(000s(0v0:'20^Th3(0[$
0r^]0n00z0,0^N00^D0x^PF0!&-1y00^ZKn_0
00(-00000>00]1800^0M0000^Y00^T0u3^U^D0:^N000a00!
0000^F:00^N^X0rP^00^A0jV00n0000^Sllp0000K00
0000^F:00^N^X0rP^00^A0jV00n0000^Sllp0000K00
^\\_00^0S^0020^0^F^T0E0owFc卐0o^B^Y0)0^Q00M^LfR0=000:N0^U00^97^R00V0^e^\\00^H0000v卐0^R0nB)00000^0S^R0T00$
bW710!n0^H5\\07
T^S00r^Q00^K^A=I0<000&0de000U^gE^0000000 00^U0000o06^e\\Y0^D#I0
R0^000;卐0iu7\\0s0^K^00meE2_00x02.00^C^AT-)0^L^G0AY^F0^LE00jG?00-:r0o0+^D5)^D7^W30p0A0卐 0^0b050_d0001^Y)^S0$
00^Bd^NN
z<^B\\00c0t^0"000
00卐 (0?x00^Qk00f00
f^Pgh900000Tf^B000|000^Z0009^P00t0^]000(2z)0^E^Z0t0n00900000k 0?0|0000)00c0^U'c^0?0
00^Y_00000^0Q 00/0jH3"z0+;00:0y0jC0000s0U;Ky^K^\\{000x0zB00q0^CS卐0L00%J00^Y000LK000Y00+_p0^QSC0800^_6^S
2^A^P4800M^H0 ex0^L0T0^L00Q0Y002^W80H0^K2E0G0%
L(^A^Q^W0j000 SK^W000 b0V0000s^L0gor)02^W0J0C0<\\.j ^EN|P^Z80hf ~^Wj00r^0?0uL0000^0^c000s^\\_00CQ^0U^0^S
00GbW^S0p0^R000M=,00!) /A0X000s^La0gor)02^W0J0C0<\\.j ^EN|P^Z80hf ~^Wj00r^0?0uL0000^0^c000s^\\_00CQ^0U^0^S
000_NdGYQ700t0RM0I^]0|00N0^CRy00^N0^S0+:f000000-4800-000:^B0j^\\^]>000so00000^Z^qD00=0\\Fp]010T^000^R0^U>$
t000Q)^DzfT+0000006u000oA0wL000-ot0^ZbV50RB\\0:f0^L800^W000Y000#00^0^N0C^0S-1000^0^E000G^Xlxt^Q0000^0T0$
0^o0j^HY0^0nP)"0,nM?FbZ3v30/!0!Xws^VA^Y0000->0rj,
00^V05^CFK00q00/00^000[G0:0^Kn^03^N0Y^B0^P00v^[00^K020000[0]'0A0uj0W %00 zj>0q[0v :00^Wuj^B

300%0F0000)0000p00)aXf00_0xq<00R"拆y0^Pff
h00
00 0y000^S
E0|000000VU0Lk00000pV~P0ZgKt00|00'^R0(>a/2'甯%RW-0c^L000^00FH890000卐 ^N<0010[y0^Vb^Sw@z00^hC):00M000(0$
000[^]S^C005000^Sr:@00j02000[E00^028)甯00000^Y0v0
#-k9e0\\=0F0H07302z^D^B) ^L^Dj000w0000 j0^b0^N^?000 卐 00b^Q00g00:^C060/W$ae00g^D00Dt?C0B/p0L00^S0j0,0003:R00$
0^F00^ZQ000u00^T00NG0\\U00^Y0/m000ZF0000;0Yj0S0^U0^00^:^F000D00300s!^W0 00'2;)B/*1r000c40t:0^00HN(0CG^000$

[ Read 62132 lines (Converted from DOS and Mac format) ]
Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Justify Where Is Next Page UnCut Text To Spell
```

Fig 2. The encrypted image inside nano, all the original bitmap bytes are completely scrambled.

```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/Task1 — ssh cybersec-server@192.168.64.6 — 122x60

cybersec-server@ubuntu:~/Desktop/Task1$ head -5 Text_file.txt
Cryptolocker is a class of malware known as ransomware, which infects a computer by restricting access to that computer's
files. Once a computer is infected, the luckless user receives a message demanding payment before files can be unlocked ag
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esktop/Task1$ openssl enc -aes-128-cbc -e -in Text_file.txt -out text_cbc.enc -K 00112233445566778899aabbccddeeff -iv 0102
030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ cat text_cbc.enc
?????9??Ru,?????#?j6?????#K?21? kAZ?^??Y?ej[?7???\]H?7k-A?K?8??*t?E??eK????b??#?wä??T.???D{2m?J?d???I???<?pU?h3???y?^??
?Lq?k?j????k?7a?0??j?X?P?W?7?7t,?????S?? ?7D????,/?K?7??7?vp?d????????????J{?????e????B? q??s\??????.Pq?%?G?7??eA?Ne?9[j
?3-??uNQ-??7GE,??-??7CT?Sdy0L?

????os:Y??n??qu?S????H[9??~?C?7B?7???<??^?
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??[?$/F?3??B?74?I1????K:~?vo?Z?u?^??b?Y?}K ??vc(k

??w??E?0???'9TWP?A????Äf????

??c??

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?/?U?F?g??CY81

??V-????????(??t?^4?7{X?b?3????f??v? ?????.n{??/[Cv????1?%?_bC?%??&?
)??k?05??r?biU!LJW_)3?P@????nc????? ?<0?`f?7?????j$5?t??r?F????i*S??EI?r.8?W?e`eD:S?J*[lM?5b?.}衛1?v??&
??)??Bj????9?AqN?5??0S
?(?44??i?%"e??")t??:]?mw%??a??k??N4D?????Y.????p?JtD??5??w?^+?HNY??j????G??v?i?J?H??i?<??I?^
?????H?d?4?+?P<
V??>DL#?3X?2a?????_T;NS[?r??"76????w<|

?*R?}??l??'/????}????g?1???G8??b#???o?S?cybersec-server@ubuntu:~/Descybersec-server@
ubuntu:~/Desktop/Task1$ openssl enc -aes-128-cbc -d -in text_cbc.enc -out text_dec.txt -K 00112233445566778899aabbccddeeff
-iv 0102030405060708090a0b0c0d0e0f10
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99aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ nano image_cbc.enc
cybersec-server@ubuntu:~/Desktop/Task1$ openssl enc -aes-128-cbc -d -in image_cbc.enc -out image_dec.bmp -K 00112233445566
778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ file image_dec.bmp
image_dec.bmp: PC bitmap, Windows 3.x format, 1658 x 1611 x 24
cybersec-server@ubuntu:~/Desktop/Task1$
```

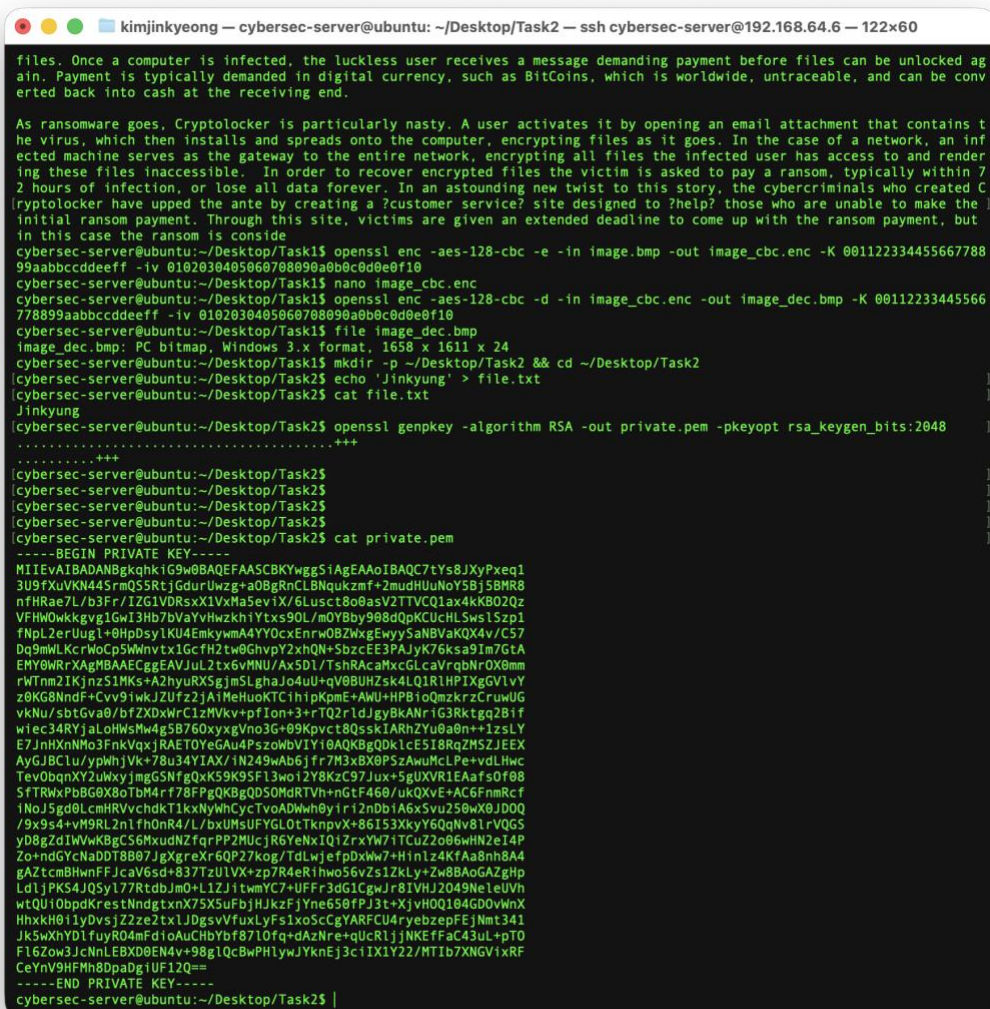
Fig 3. The full round trip for both file - after decrypt, head text_dec.txt show the original text again, and file image_dec.bmp say "PC bitmap ... 1658 x 1611", so the decryption bring back the files exactly the same.

Observation: when you encrypt a file with some cipher/mode it give a totally different unreadable output, and you need the same key + IV to reverse it. Also the mode you pick is actually matter a lot in the real world - ECB mode leak the structure because the same plaintext block always turn into the same ciphertext block, but CBC (the one i used) chain the blocks together with a IV so no pattern is showing. This is basically the exact weakness behind the Adobe breach in 2013, they stored around 153 million password with 3DES in ECB mode and the researchers manage to recover the common ones without even breaking the key at all.

Task 2 – Asymmetric (RSA) Encryption

Asymmetric encryption is when there is a key pair - a public key that anyone can use for encrypt, and a private key which only the owner have for decrypt. So i made a file with my name inside, generate the key pair, and then encrypt using the public key and decrypt using the private one.

```
echo 'Jinkyung' > file.txt openssl genpkey -algorithm RSA -out private.pem -pkeyopt  
rsa_keygen_bits:2048 openssl rsa -pubout -in private.pem -out public.pem openssl rsautl -  
encrypt -inkey public.pem -pubin -in file.txt -out file.enc openssl rsautl -decrypt -inkey  
private.pem -in file.enc -out dec.txt
```



```
kimjinkyung — cybersec-server@ubuntu: ~/Desktop/Task2 — ssh cybersec-server@192.168.64.6 — 122x60

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cybersec-server@ubuntu:~/Desktop/Task1$ openssl enc -aes-128-cbc -e -in image.bmp -out image_cbc.enc -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ nano image_cbc.enc
cybersec-server@ubuntu:~/Desktop/Task1$ openssl enc -aes-128-cbc -d -in image_cbc.enc -out image_dec.bmp -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
cybersec-server@ubuntu:~/Desktop/Task1$ file image_dec.bmp
image_dec.bmp: PC bitmap, Windows 3.x format, 1658 x 1611 x 24
cybersec-server@ubuntu:~/Desktop/Task1$ mkdir -p ~/Desktop/Task2 && cd ~/Desktop/Task2
cybersec-server@ubuntu:~/Desktop/Task2$ echo 'Jinkyung' > file.txt
cybersec-server@ubuntu:~/Desktop/Task2$ cat file.txt
Jinkyung
cybersec-server@ubuntu:~/Desktop/Task2$ openssl genpkey -algorithm RSA -out private.pem -pkeyopt rsa_keygen_bits:2048
.....+++
.....+++
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$ cat private.pem
-----BEGIN PRIVATE KEY-----
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBAQwggSiAgEAAoIBAQC7tYs8JXyPxeq1
3U9fxuVKN445rmQSSrTjGdurUwzg+a0BgRnCLBNqukmf+2mudHUuNoY5Bj58MR8
nfhRae7L/b3Fr/IZG1VDRsxXVxMaSeviX/6Lusct8o0asV2TTCVQ1ax4kKB02Qz
VFHW0wkkgvg1GwI3Hb7bVaYVhwzkhYtXs90L/mOYBby908dQpKCUCiHLSws1Szp1
fNpL2erUugl+0HpDsyLkU4EmkywmA4YY0cxEnrW0BZwxEwySaNBVaQX4v/C57
Dq9mWLKcrWoCp5Wwvnt1GcfH2tw0GhvpY2xhQn+SbzcEE3PAJyK76ksa9Im7GtA
EMY0WRrXAgMBAEAggEAVJUL2tx6vMNU/Ax5D1/TshRacaMxGCLcaVrbqNrOX0mm
rWTnm2IKjnz51MKS+A2hyuRXSgjmSLghaJo4uU+qV0BUH2sk4LQ1R1HPiXGv1vY
z0KG8NndF+Cvv9iWkJ2Uf2zjAiMeHuokTCihipKpmE+AMU+HPBiOQmzkrzCruwUG
vKnu/sbtGva0/bfZXDxWrc1zMKv+pfIon+3+rTQr1dJgyBkAnriG3Rktgq2BiF
wIec34RYjaLoHwMw4g5B760xyxgVno3G+09KpvcT8QsskIARhZY0a0n++1zsLY
E7JnHxNMo3FnkVqxjRAET0YeGau4Psz0w0V1Y0AQKbQDk1cE5I8RqZMSZJEEX
AyGJBCLu/yphWjVjK+78u34YIAX/iN249wAb6jfr7M3xBX0PSzAwuMcLPe+vdLHwc
Tev0bqnXY2UwxyjmgGSNfgQxK59K95F13woi2Y8KzC97Jux+5gUXVR1EAafs0f08
SfTRWxPbB60X80TbM4rf78FPgQKbQD50MdrTVh+nGTF460/ukQXvE+AC6FnmRcf
iNoJ5gd0LcmHRVvchdkT1kxNpYWhCycTvoADMWh0yiri2nDbiA6xSvu250wX0JD0Q
/9x9s4+vM9RL2n1fhOnR4/L/bxUmsUFYGL0TtknpvX+86I53XkyY6QqNv81rVQGS
yD8gZdIWwKBGCS6MxudNZfqrPP2MucjR6YeNxIQ1ZrxYw7iTCuZ2o06wHN2eI4P
Zo+ndGyCNaDDT8B07JgXgreXr6QP27kog/TdLwjeFpDxwW7+HInLz4KfAa8nh8A4
gAZtcmBhwnFFJcaV6sd+837ZtU1VX+zp74RerIhwo56vZs1ZkLy+Zw8BA0GAZGHP
Ld1jPKS4JQSy177RtdbJm0+L1ZJ1twmY7+UFFr3dG1CgwJr8IVHJ2049Ne1eUVh
wtQ10bpdKrestNndgtxn75X5uFbJHJkZfjYne650FPJ3t+XjvHQ104G00vWnX
HhxxH01lyDvsjZ2ze2tLJJDgsVfuxLyF51xoScCYARFCU4ryebzefFEjNmt341
JK5wXhYD1fuyR04MfdioAuChbYbF8710fq+dAZNre+qUR1jJNKEfFaC43uL+pTO
F16Zow3JcNnLEBXD0EN4v+98g1QcBwPH1YwJYknEj3ciIX1Y22/MTIb7XNGVixRF
CeYnV9HFHh8DpaDg1UF12Q==
-----END PRIVATE KEY-----
cybersec-server@ubuntu:~/Desktop/Task2$ |
```

Fig 4. (1) file.txt with my name inside, and (2) the private key (private.pem) that got generated.

```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/Task2 — ssh cybersec-server@192.168.64.6 — 122x60

rWfm2IKjnzS1MKs+A2hyuRXSgjsLghaJo4uU+qV0BUH2sk4LQ1RLHPiXgVlVY
z0KG8NndF+Cvv9iWkZJufz2jA1MeHuokTCih1pKpME+AHU+HPB1oQmzkrzCruwUG
vkNu/sbtGva0/bfZXDXwC1zWkv+pfIon+3+rTQ2rldJgyBKAhr1G3Rktgg2B1f
w1ec34RYja1oHw5Mw4g5B760xyxgVno3C+09Kpvt80ssk1ARhZYu0a0n++1zsLY
E7JnHxNMo3FnkVqxjRAET0YeGau4PszowbVIY10AQK8gQDk1cE518RqZMSZIEEX
AyGJBCLu/yphWjVik+78u34YIAX/1N249wAb6jfr7M3XB0PSZAwuMcLPe+vdLHwc
TevObqnXY2UkxyjmgGSNfgQxK59K9Sf13wo12Y8KzC97Jux+5glUXVR1EAfs0f08
SfTRwxPbB8G0X8oTbM4r+f78FPgQK8gQD50MdrTVh+nGtf460/ukQXvE+AC6FnmRcf
iNoJ5gd0LcmHRVvchdKt1kxNyhWycTvoADWwh0y1r12nDb1A6Svu250wX0JDOQ
/9x9s4+vm9RL2n1fhOnR4/L/bxUMsUFYGL0tTknpxX+86153XkyY6QqNv81rVQGS
yD8gzd1WwK8gC56MxudNZfqrPP2MUCjR6YeN1Q1ZrxYW71TCuZ2o06wHN2e14P
Zo+ndGyCNaDDT8B07JgXgreXr60P27kog/TdLwjeFpDxWw7+Hin1z4KfAa8nh8A4
gAZtcmBHwnFFJcaV6sd+837TzU1VX+zp7R4eR1hwo56vZs1ZKLy+Zw8BAoGAZgHp
LdljPKS4JQsYl77RtdbJm0+L1ZJ1tWmYC7+UFFr3dG1CgwJr81VHJ2049Ne1UVh
wtQU10bpdKrestNndgtxnX75SuFbjHJkzFjYne650fP3jt+XjvHOQ104G00vWnX
HhXkH01lyDvsj22ze2tx1JdgsVfVuxLyFs1xoScCYARFCU4ryebzepFEjNmt341
Jk5WthYD1fuyR04mfDioAuChbYbf8710fq+dAzNre+qUCR1jNKEfFaC43uL+pT0
F16Zow3JcNnLEBX08EN4+98glQcBwPhlyJYknEj3ciIX1Y22/MT1b7XNGVixRF
CeYnV9HFHh8DpaDgUFI12Q==
-----END PRIVATE KEY-----
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout 0in private.pem -out public.pem
unknown option 0in
rsa [options] <infile> >outfile
where options are
  -inform arg      input format - one of DER NET PEM
  -outform arg     output format - one of DER NET PEM
  -in arg          input file
  -sgckey          Use IIS SGC key format
  -passin arg      input file pass phrase source
  -out arg         output file
  -passout arg     output file pass phrase source
  -des             encrypt PEM output with cbc des
  -des3            encrypt PEM output with ede cbc des using 168 bit key
  -seed           encrypt PEM output with cbc seed
  -aes128, -aes192, -aes256
                  encrypt PEM output with cbc aes
  -camellia128, -camellia192, -camellia256
                  encrypt PEM output with cbc camellia
  -text           print the key in text
  -noout         don't print key out
  -modulus       print the RSA key modulus
  -check         verify key consistency
  -pubin         expect a public key in input file
  -pubout        output a public key
  -engine e      use engine e, possibly a hardware device.
[cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
cat: public.pem: No such file or directory
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout -in private.pem -out public.pem
writing RSA key
[cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAu7WLPcV8j8Xqtd1PX17L
Sje0Eq5kEuUbyXnbq1MM4PmjgYEZwiwTarpM5n/tprnR1LjaG0QY+QTEfJ3x0Wnu
y/29xa/yGRtVQ0bMV9VcTGuXr41/+17rHLfKNGrFdk01QkNWseJCgTtkM1RR1jsJ
JIL4NRsCNx2+21WmLx8MSIYmLcbPT1/5jmaW8vdPHUKSg1HBy0sLJUs6dXzaS9nq
1LoJftB6Q7MpS10BJpMsJgOGGDnMRJ68DgWVsYBMMskmjQVW1kF+L/wuew6vZ1iy
nK1qAqeV1p77cdRnHx9rcNBo06WNSYUDfkm83BBNzWcc1u+pLGvSjuxrQB0GNFka
1wIDAQAB
-----END PUBLIC KEY-----
[cybersec-server@ubuntu:~/Desktop/Task2$
```

Fig 5. (3) the public key (public.pem), which is derived from the private key.

```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/Task2 — ssh cybersec-server@192.168.64.6 — 122x60

yD8gZd1WwKbgCS6MxudNZfqrPP2MucjR6YeNxiQ1ZrxYW71TCuZ2o86wHN2e14P
Zo+ndGYcNaDDT8B07JgXgreXr60P27kog/TdLwjeFpDxWw7+H1nlz4KfAa8nh8A4
gAZtcmBHwnFFJcaV6sd+837TzU1VX+zp7R4eRihwo56vZs1ZKLy+Zw8BAoGAZgHp
LdljPKS4JQsYl77RtdbJm0+L1ZJitwYc7+UFFr3dG1CgwJr8IVHJ2049NeleUVh
wtQU10bpdKrestNndgtxnX75XSuFbjHJkzFjYne650fPJ3t+XjvHQ104GD0vWnX
HhXkH01lyDvsjZ2ze2txlJDgsVvFuxLyFs1xoScGyARFCU4ryebzepFEjNmt341
Jk5WthYD1fuyR04mfDioAuChbYbf8710fq+dazNre+qUcR1jjNKEffac43uL+pT0
F16Zow3JcNnLEBXD0EN4v+98glQcBwPHlywJYknEj3ciIX1Y22/MT1b7XNGVixRF
CeYnV9HFHh8DpadgiUFI2Q==
-----END PRIVATE KEY-----
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout 0in private.pem -out public.pem
unknown option 0in
rsa [options] <infile> >outfile
where options are
  -inform arg      input format - one of DER NET PEM
  -outform arg     output format - one of DER NET PEM
  -in arg          input file
  -sgckey          Use IIS SGC key format
  -passin arg      input file pass phrase source
  -out arg         output file
  -passout arg     output file pass phrase source
  -des            encrypt PEM output with cbc des
  -des3           encrypt PEM output with ede cbc des using 168 bit key
  -seed           encrypt PEM output with cbc seed
  -aes128, -aes192, -aes256
                  encrypt PEM output with cbc aes
  -camellia128, -camellia192, -camellia256
                  encrypt PEM output with cbc camellia
  -text           print the key in text
  -noout          don't print key out
  -modulus        print the RSA key modulus
  -check          verify key consistency
  -pubin          expect a public key in input file
  -pubout         output a public key
  -engine e       use engine e, possibly a hardware device.
cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
cat: public.pem: No such file or directory
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout -in private.pem -out public.pem
writing RSA key
cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAu7WLPcV8j8Xqtd1PX17l
Sje0Eq5kEuUbyXnbq1MM4PmjgYEZwiwTarpM5n/tprnR1LjaG0QY+QTEfJ3x0Wnu
y/29xa/yGRtVQ0bMV9VcTGuXr4l/+17rHLfKNGrFdk01QkNmSeJCgTtkM1RR1jsJ
J1L4NRsCNx2+21WmLx8MSIYmLcbPT1/5jmaW8vdPHUKSg1HBy0sLJUs6dXzaS9nq
1LoJftB6Q7Mp51OBjPmSJgOGGdnMRJ68DgwVsYBMmskmjQVW1kF+L/wuew6vZ1iy
nK1qAqeVlp77cdRnHx9rcNB0b6WnsYUDfkm83BBNzwCc1u+pLGVsJuxrQB0GNFka
1wIDAQAB
-----END PUBLIC KEY-----
cybersec-server@ubuntu:~/Desktop/Task2$ /Users/Kimjinkyong/Desktop/Screenshot\ 2026-08-24\ at\ 6.55.16 pm.png ^C
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsautl -encrypt -inkey public.pem -pubin -in file.txt -out file.enc
cyb7os?????1Y??r7ap"?%? ?sktop/Task2$ cat file.enc
Y7d?-??g??cm????/?-97T??F.??-?p?v?????&r2E)?4?a=???^?      ???5?Z78-vj?z??=?0?Z`oRL???敲??QK?????f?u5'h??xC?
vr??E?Z?}??????*0c(7L676L?ytrL)AX???
S-?
YE|??YJ?_oP:7L??X?v>??cybersec-server@ubuntu:~/Desktop/Task2$
```

Fig 6. (4) encrypting with the public key, then cat file.enc show the scrambled ciphertext.

```

kimjinkyong — cybersec-server@ubuntu: ~/Desktop/Task2 — ssh cybersec-server@192.168.64.6 — 122x60
wtQUi0bpdKrestNndgtxnX75X5uFbjHJkzFjYne650fPJ3t+XjvHQ104GD0vWnX
HhXkH01yDvsj22ze2txlJDgsVfuxLyFs1xo5CcYARFCU4ryebzepFEjNmt341
Jk5wXhYD1fuyR04mfDioAuChbYbf8710fq+dazNre+qUcR1jNKEfFAC43uL+pT0
F16Zow3JcNnLEBXD0EN4v+98glQcBwPHlywJYknEj3ciIX1Y22/MT1b7XNGVixRF
CeYnV9HfMh8DpaDgiUF12Q==
-----END PRIVATE KEY-----
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout 0in private.pem -out public.pem
unknown option 0in
rsa [options] <infile >outfile
where options are
  -inform arg      input format - one of DER NET PEM
  -outform arg     output format - one of DER NET PEM
  -in arg          input file
  -sgckey          Use IIS SGC key format
  -passin arg      input file pass phrase source
  -out arg         output file
  -passout arg     output file pass phrase source
  -des            encrypt PEM output with cbc des
  -des3           encrypt PEM output with ede cbc des using 168 bit key
  -seed           encrypt PEM output with cbc seed
  -aes128, -aes192, -aes256
                    encrypt PEM output with cbc aes
  -camellia128, -camellia192, -camellia256
                    encrypt PEM output with cbc camellia
  -text           print the key in text
  -noout          don't print key out
  -modulus        print the RSA key modulus
  -check          verify key consistency
  -pubin          expect a public key in input file
  -pubout         output a public key
  -engine e       use engine e, possibly a hardware device.
[cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
cat: public.pem: No such file or directory
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout -in private.pem -out public.pem
writing RSA key
[cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBcGKCAQEAu7WLPcV8j8Xqtd1PX17l
SjeOEg5kEuUbyXnbq1MM4PmjgYEZwiwTarpM5n/tprnR1LjaG0QY+QTEfJ3x0Wnu
y/29xa/yGRtVQ0bMV9VcTGuXr4l/+i7rHLfKNGrFdk01QkNWseJCgTtkM1RR1jsJ
J1L4NRsCNx2+21WmLx8M5IYmLcbPTi/5jmAW8vdPHUKSg1HBy0sLJUs6dXzaS9nq
1LoJftB6Q7Mp51OBjPmSjgOGGdnMRJ68DgwVsYBMmskmjQVWkF+L/wuew6vZ1iy
nK1qAqeVlp77cdRnHx9rcNB0b6WnsYUDfkm83BBNzwcC1u+pLGv5JuxrQB0GNFka
1wIDAQAB
-----END PUBLIC KEY-----
[cybersec-server@ubuntu:~/Desktop/Task2$ /Users/kimjinkyong/Desktop/Screenshot\ 2026-08-24\ at\ 6.55.16 pm.png ^C
[cybersec-server@ubuntu:~/Desktop/Task2$
[cybersec-server@ubuntu:~/Desktop/Task2$
[cybersec-server@ubuntu:~/Desktop/Task2$
[cybersec-server@ubuntu:~/Desktop/Task2$
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsautl -encrypt -inkey public.pem -pubin -in file.txt -out file.enc
cyb7os?????1Y??r7ap"?%? ?sktop/Task2$ cat file.enc
Y7d7-??g?cm?????/?-9?T?F?z??p?Vv?????&r2E)?4?a=?^? ???5?Z78-vj?z???0?Z'oRL???截??QK????? ?????f?u5'?h??x?C?
\r7z?Z7Z)??????2*0c(7L676LytrL)AX????
S-?
YEj??YJ? oP:7L??X?v>??cybersec-server@ubuntu:~/Desktop/Task2$
[cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsautl -decrypt -inkey private.pem -in file.enc -out dec.txt
Jinkyung
[cybersec-server@ubuntu:~/Desktop/Task2$ ]

```

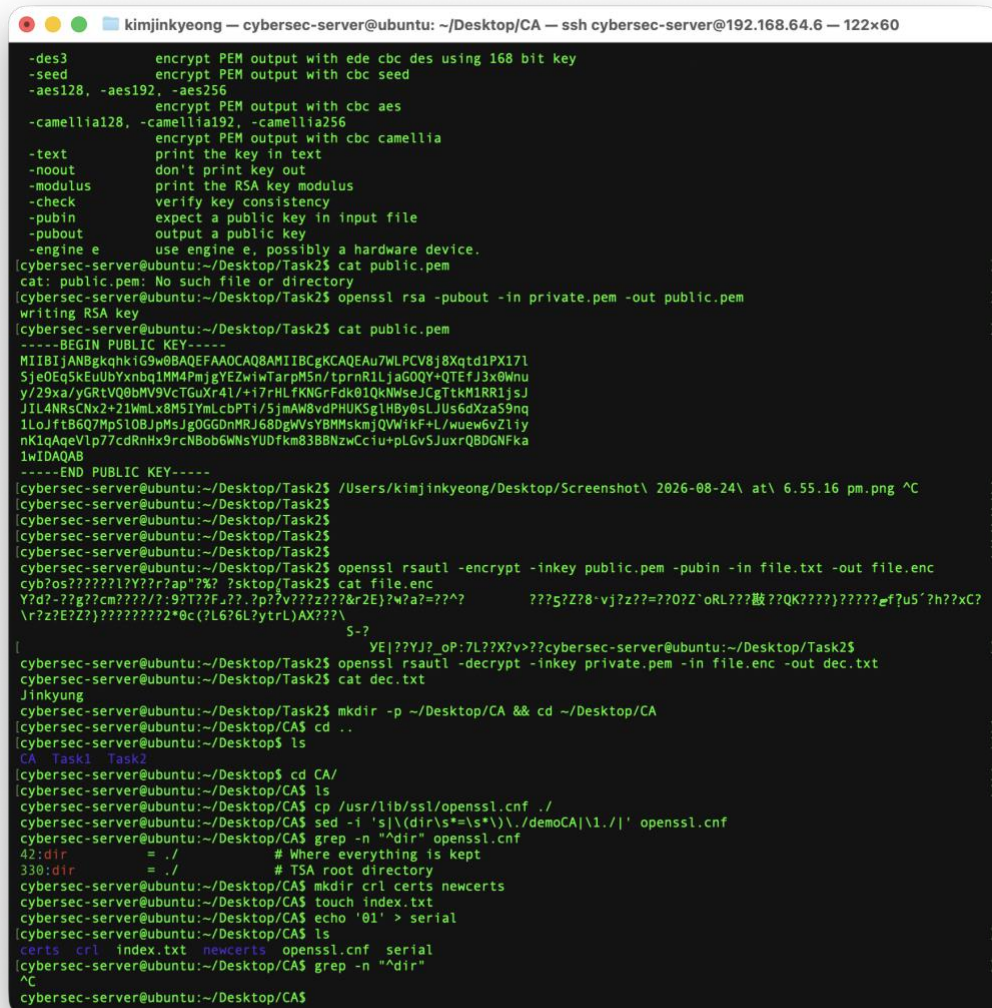
Fig 7. (5) decrypting with the private key - cat dec.txt give back my name, so only the private key can undo what the public key locked.

Observation: whatever you encrypt with the public key can only be open with the matching private key. Which mean the private key is like the most sensitive thing in the whole scheme - the Heartbleed bug (2014) is a good example here, it was a memory leak flaw in OpenSSL that let the attacker read the server private key straight out of the memory, so they could decrypt the traffic or pretend to be the server, doesn't matter how strong RSA itself is.

Task 3 – Becoming a Certificate Authority (CA)

In this task i make myself into a root CA. First i copy the OpenSSL config file, change its working directory to the current folder, and create the folder/file structure that the CA database is needing. After that i generate a self-signed root certificate.

```
cp /usr/lib/ssl/openssl.cnf ./ # edited openssl.cnf so that: dir = ./ mkdir crl certs newcerts
touch index.txt echo '01' > serial openssl req -new -x509 -keyout ca.key -out ca.crt -config
openssl.cnf
```



```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/CA — ssh cybersec-server@192.168.64.6 — 122x60
-des3          encrypt PEM output with ede cbc des using 168 bit key
-seed          encrypt PEM output with cbc seed
-aes128, -aes192, -aes256  encrypt PEM output with cbc aes
-camellia128, -camellia192, -camellia256  encrypt PEM output with cbc camellia
-text          print the key in text
-noout         don't print key out
-modulus       print the RSA key modulus
-check         verify key consistency
-pubin         expect a public key in input file
-pubout        output a public key
-engine e      use engine e, possibly a hardware device.
cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
cat: public.pem: No such file or directory
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsa -pubout -in private.pem -out public.pem
writing RSA key
cybersec-server@ubuntu:~/Desktop/Task2$ cat public.pem
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAu7WLPV8j8Xqtd1PX17l
SjeOE5kEulbYxnbq1MM4PmjgYEZw1wTarpM5n/tpnrR1LjaG0QY+QTEfJ3x0Wnu
y/29xa/yGR1VQ0bMV9VcTGUXr4l/+17rHLfKNGrFdk01QkNmseJGtTtkM1RR1jsJ
J1L4NRsCNx2+21Wml.x8M5IYml.cbPTI/5jMAW8vdPHUKSg1HBy0sLJUs6dXzaS9nq
1LoJftB6Q7Mp51OB.jpMsJgOGGdnMRJ68DgWVsYBMmskmjQVW1kF+L/wuew6vZliy
nK1qAqeVlp77cdRnHx9rcNBob6WnsYUDfkm83BBNzwcC1u+plGv5JuxrQB0GNFka
1wIDAQAB
-----END PUBLIC KEY-----
cybersec-server@ubuntu:~/Desktop/Task2$ /Users/kimjinkyong/Desktop/Screenshot\ 2026-08-24\ at\ 6.55.16 pm.png ^C
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsautl -encrypt -inkey public.pem -pubin -in file.txt -out file.enc
cyb7os?????lY??r7ap"?%? ?sktop/Task2$ cat file.enc
Y7d?-??g??cm??????:92T??Fj?? ?p?v????z??&r2E)?4?a=?^?      ???5?Z78-vj?z??=?0?Z"oRL???敲??QK????)?????ef?u5'?h??xC?
Vr??Z7Z??)??????2?0c(?L67L?ytrL)AX???)
S-?
YE)??YJ?_oP:7L?X?V>??cybersec-server@ubuntu:~/Desktop/Task2$
cybersec-server@ubuntu:~/Desktop/Task2$ openssl rsautl -decrypt -inkey private.pem -in file.enc -out dec.txt
cybersec-server@ubuntu:~/Desktop/Task2$ cat dec.txt
Jinkyung
cybersec-server@ubuntu:~/Desktop/Task2$ mkdir -p ~/Desktop/CA && cd ~/Desktop/CA
cybersec-server@ubuntu:~/Desktop/CA$ cd ..
cybersec-server@ubuntu:~/Desktop$ ls
CA Task1 Task2
cybersec-server@ubuntu:~/Desktop$ cd CA/
cybersec-server@ubuntu:~/Desktop/CA$ ls
cybersec-server@ubuntu:~/Desktop/CA$ cp /usr/lib/ssl/openssl.cnf ./
cybersec-server@ubuntu:~/Desktop/CA$ sed -i 's|\\(dir\\s*=\\s*\\)\\.\\.demoCA|\\1\\.\\.|' openssl.cnf
cybersec-server@ubuntu:~/Desktop/CA$ grep -n "dir" openssl.cnf
42:dir          = ./          # Where everything is kept
330:dir         = ./          # TSA root directory
cybersec-server@ubuntu:~/Desktop/CA$ mkdir crl certs newcerts
cybersec-server@ubuntu:~/Desktop/CA$ touch index.txt
cybersec-server@ubuntu:~/Desktop/CA$ echo '01' > serial
cybersec-server@ubuntu:~/Desktop/CA$ ls
certs crl index.txt newcerts openssl.cnf serial
cybersec-server@ubuntu:~/Desktop/CA$ grep -n "dir"
^C
cybersec-server@ubuntu:~/Desktop/CA$
```

Fig 8. The dir in the config is now ./, and ls show the CA structure: certs, crl, index.txt, newcerts, openssl.cnf, serial.

```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/CA — ssh cybersec-server@192.168.64.6 — 122x60
cybersec-server@ubuntu:~/Desktop/CA$ cp /usr/lib/ssl/openssl.cnf ./
cybersec-server@ubuntu:~/Desktop/CA$ sed -i 's|(dir\s*=\s*\)|demoCA|1.1|' openssl.cnf
cybersec-server@ubuntu:~/Desktop/CA$ grep -n "^dir" openssl.cnf
42:dir          = ./                                # Where everything is kept
330:dir         = ./                                # TSA root directory
cybersec-server@ubuntu:~/Desktop/CA$ mkdir crl certs newcerts
cybersec-server@ubuntu:~/Desktop/CA$ touch index.txt
cybersec-server@ubuntu:~/Desktop/CA$ echo '01' > serial
cybersec-server@ubuntu:~/Desktop/CA$ ls
certs  crl  index.txt  newcerts  openssl.cnf  serial
cybersec-server@ubuntu:~/Desktop/CA$ grep -n "^dir"
^C
cybersec-server@ubuntu:~/Desktop/CA$ openssl req -new -x509 -keyout ca.key -out ca.crt -config openssl.cnf
Generating a 2048 bit RSA private key
.....+++
writing new private key to 'ca.key'
[Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
Verify failure
[Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
[Country Name (2 letter code) [AU]:AU
[State or Province Name (full name) [Some-State]:NSW
[Locality Name (eg, city) []:SYD
[Organization Name (eg, company) [Internet Widgits Pty Ltd]:FEIT
[Organizational Unit Name (eg, section) []:^C
cybersec-server@ubuntu:~/Desktop/CA$ openssl req -new -x509 -keyout ca.key -out ca.crt -config openssl.cnf
Generating a 2048 bit RSA private key
.....+++
writing new private key to 'ca.key'
[Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
[Country Name (2 letter code) [AU]:AU
[State or Province Name (full name) [Some-State]:NSW
[Locality Name (eg, city) []:SYD
[Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
[Organizational Unit Name (eg, section) []:FEIT
[Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
[Email Address []:root@cybersec.com.au
cybersec-server@ubuntu:~/Desktop/CA$ ls
ca.crt  ca.key  certs  crl  index.txt  newcerts  openssl.cnf  serial
cybersec-server@ubuntu:~/Desktop/CA$
```

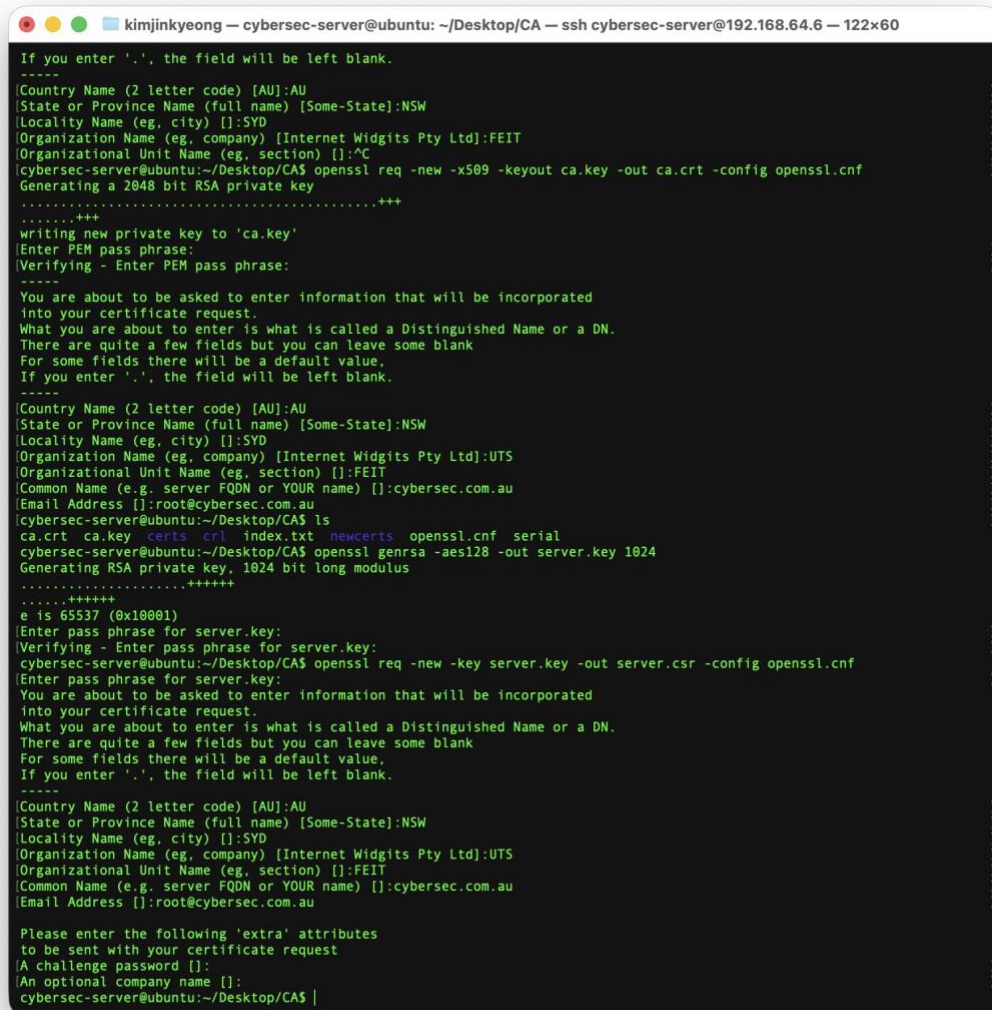
Fig 9. Generating the self-signed CA cert. The details i put: AU / NSW / SYD / UTS / FEIT / cybersec.com.au / root@cybersec.com.au. After that ls show ca.key and ca.crt is created.

Observation: the root CA private key (ca.key) is very powerful - whoever have it can sign a "valid" certificate for basically any name. This is exactly what happen in the DigiNotar (2011) case, the attacker used the trusted CA own signing process to issue more than 500 fake certificates (one of them was even for *.google.com), and that is the reason why the real CAs are audited so heavily.

Task 4 – Issuing an X.509 Certificate for cybersec.com.au

Now i am acting as the CA and i issue a certificate to a "website". There is 3 steps for this one - generate the server key, make a Certificate Signing Request (CSR), and then let the CA sign it.

```
openssl genrsa -aes128 -out server.key 1024
openssl req -new -key server.key -out server.csr -config openssl.cnf
openssl ca -in server.csr -out server.crt -cert ca.crt -keyfile ca.key -config openssl.cnf
```



```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/CA — ssh cybersec-server@192.168.64.6 — 122x60
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:SYD
Organization Name (eg, company) [Internet Widgits Pty Ltd]:FEIT
Organizational Unit Name (eg, section) []:^C
[cybersec-server@ubuntu:~/Desktop/CA$ openssl req -new -x509 -keyout ca.key -out ca.crt -config openssl.cnf
Generating a 2048 bit RSA private key
+++++
writing new private key to 'ca.key'
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:SYD
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
Organizational Unit Name (eg, section) []:FEIT
Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
Email Address []:root@cybersec.com.au
[cybersec-server@ubuntu:~/Desktop/CA$ ls
ca.crt ca.key certs csl index.txt newcerts openssl.cnf serial
cybersec-server@ubuntu:~/Desktop/CA$ openssl genrsa -aes128 -out server.key 1024
Generating RSA private key, 1024 bit long modulus
+++++
e is 65537 (0x10001)
Enter pass phrase for server.key:
Verifying - Enter pass phrase for server.key:
cybersec-server@ubuntu:~/Desktop/CA$ openssl req -new -key server.key -out server.csr -config openssl.cnf
Enter pass phrase for server.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:SYD
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
Organizational Unit Name (eg, section) []:FEIT
Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
Email Address []:root@cybersec.com.au

Please enter the following 'extra' attributes
to be sent with your certificate request
[A challenge password []:
[An optional company name []:
cybersec-server@ubuntu:~/Desktop/CA$ ]
```

Fig 10. Generating the server key and the CSR command with the details i entered (Organization = UTS, Common Name = cybersec.com.au). The challenge password and the optional company name i left them blank.

```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/CA — ssh cybersec-server@192.168.64.6 — 122x60
cybersec-server@ubuntu:~/Desktop/CA$ openssl req -new -key server.key -out server.csr -config openssl.cnf
Enter pass phrase for server.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:SYD
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
Organizational Unit Name (eg, section) []:FEIT
Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
Email Address []:root@cybersec.com.au

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
cybersec-server@ubuntu:~/Desktop/CA$ openssl ca -in server.csr -out server.crt -cert ca.crt -keyfile ca.key -config openss
l.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
    Serial Number: 1 (0x1)
    Validity
        Not Before: Aug 24 09:16:04 2026 GMT
        Not After : Aug 24 09:16:04 2027 GMT
    Subject:
        countryName           = AU
        stateOrProvinceName   = NSW
        organizationName      = UTS
        organizationalUnitName = FEIT
        commonName            = cybersec.com.au
        emailAddress          = root@cybersec.com.au
    X509v3 extensions:
        X509v3 Basic Constraints:
            CA:FALSE
        Netscape Comment:
            OpenSSL Generated Certificate
        X509v3 Subject Key Identifier:
            7F:4D:15:1F:3B:29:28:0A:08:A3:52:36:57:40:24:BC:FB:6E:FE:4D
        X509v3 Authority Key Identifier:
            keyid:08:EB:5E:70:AD:42:D9:22:8C:DF:08:80:75:2A:4B:4D:5F:FC:A7:4A
    Certificate is to be certified until Aug 24 09:16:04 2027 GMT (365 days)
    Sign the certificate? [y/n]:y

1 out of 1 certificate requests certified, commit? [y/n]:y
Write out database with 1 new entries
Data Base Updated
cybersec-server@ubuntu:~/Desktop/CA$ ls
ca.crt  certs  index.txt      index.txt.old  openssl.cnf  serial.old  server.csr
ca.key  crl    index.txt.attr newcerts       serial       server.crt  server.key
cybersec-server@ubuntu:~/Desktop/CA$
```

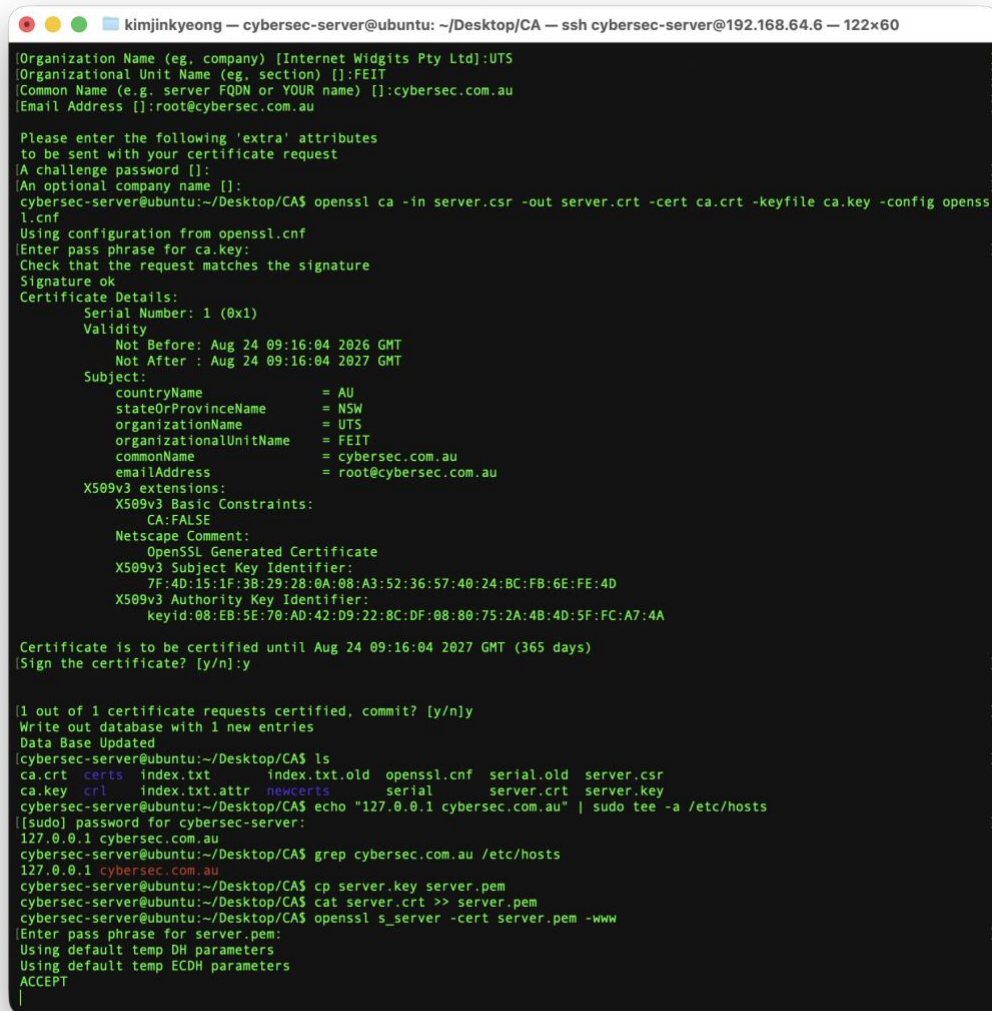
Fig 11. The CA signing the CSR - "Signature ok", the certificate details, then **Sign? y / commit? y** and "Data Base Updated". Now ls show the issued server.crt.

Observation: the openssl ca step is the exact legit signing operation that issue a trusted certificate - which is the same step the DigiNotar attackers abused from inside the compromised CA to issue certificates for domain they dont even own.

Task 5 – Using PKI for a Web Site (HTTPS)

For the last task i use the certificate to run a HTTPS server and see how the browser trust is actually working. I point the domain to localhost, put the key and the certificate together into one server.pem, and then launch the OpenSSL test web server.

```
echo "127.0.0.1 cybersec.com.au" | sudo tee -a /etc/hosts && cp server.key server.pem && cat server.crt >> server.pem && openssl s_server -cert server.pem -www
```



```
kimjinkyong — cybersec-server@ubuntu: ~/Desktop/CA — ssh cybersec-server@192.168.64.6 — 122x60
[Organization Name (eg. company) [Internet Widgits Pty Ltd]:UTS
[Organizational Unit Name (eg. section) []:FEIT
[Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
[Email Address []:root@cybersec.com.au

Please enter the following 'extra' attributes
to be sent with your certificate request
[A challenge password []:
[An optional company name []:
cybersec-server@ubuntu:~/Desktop/CA$ openssl ca -in server.csr -out server.crt -cert ca.crt -keyfile ca.key -config openss
l.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
  Serial Number: 1 (0x1)
  Validity
    Not Before: Aug 24 09:16:04 2026 GMT
    Not After : Aug 24 09:16:04 2027 GMT
  Subject:
    countryName           = AU
    stateOrProvinceName   = NSW
    organizationName       = UTS
    organizationalUnitName = FEIT
    commonName             = cybersec.com.au
    emailAddress           = root@cybersec.com.au
  X509v3 extensions:
    X509v3 Basic Constraints:
      CA:FALSE
    Netscape Comment:
      OpenSSL Generated Certificate
    X509v3 Subject Key Identifier:
      7F:4D:15:1F:38:29:28:0A:08:A3:52:36:57:40:24:8C:FB:6E:FE:4D
    X509v3 Authority Key Identifier:
      keyid:08:EB:5E:70:AD:42:D9:22:8C:DF:08:80:75:2A:4B:4D:5F:FC:A7:4A

Certificate is to be certified until Aug 24 09:16:04 2027 GMT (365 days)
[Sign the certificate? [y/n]:y

[1 out of 1 certificate requests certified, commit? [y/n]:y
Write out database with 1 new entries
Data Base Updated
cybersec-server@ubuntu:~/Desktop/CA$ ls
ca.crt  certs  index.txt  index.txt.old  openssl.cnf  serial.old  server.csr
ca.key  crl    index.txt.attr  newcerts      serial       server.crt  server.key
cybersec-server@ubuntu:~/Desktop/CA$ echo "127.0.0.1 cybersec.com.au" | sudo tee -a /etc/hosts
127.0.0.1 cybersec.com.au
cybersec-server@ubuntu:~/Desktop/CA$ grep cybersec.com.au /etc/hosts
127.0.0.1 cybersec.com.au
cybersec-server@ubuntu:~/Desktop/CA$ cp server.key server.pem
cybersec-server@ubuntu:~/Desktop/CA$ cat server.crt >> server.pem
cybersec-server@ubuntu:~/Desktop/CA$ openssl s_server -cert server.pem -www
Enter pass phrase for server.pem:
Using default temp DH parameters
Using default temp ECDH parameters
ACCEPT
```

Fig 12. Launching the HTTPS server with server.pem - it reach the ACCEPT state and listen on port 4433.

Before trusting the CA

When i first visit <https://cybersec.com.au:4433> in Firefox it give me a warning, because the browser have no reason yet to trust my self-signed CA:

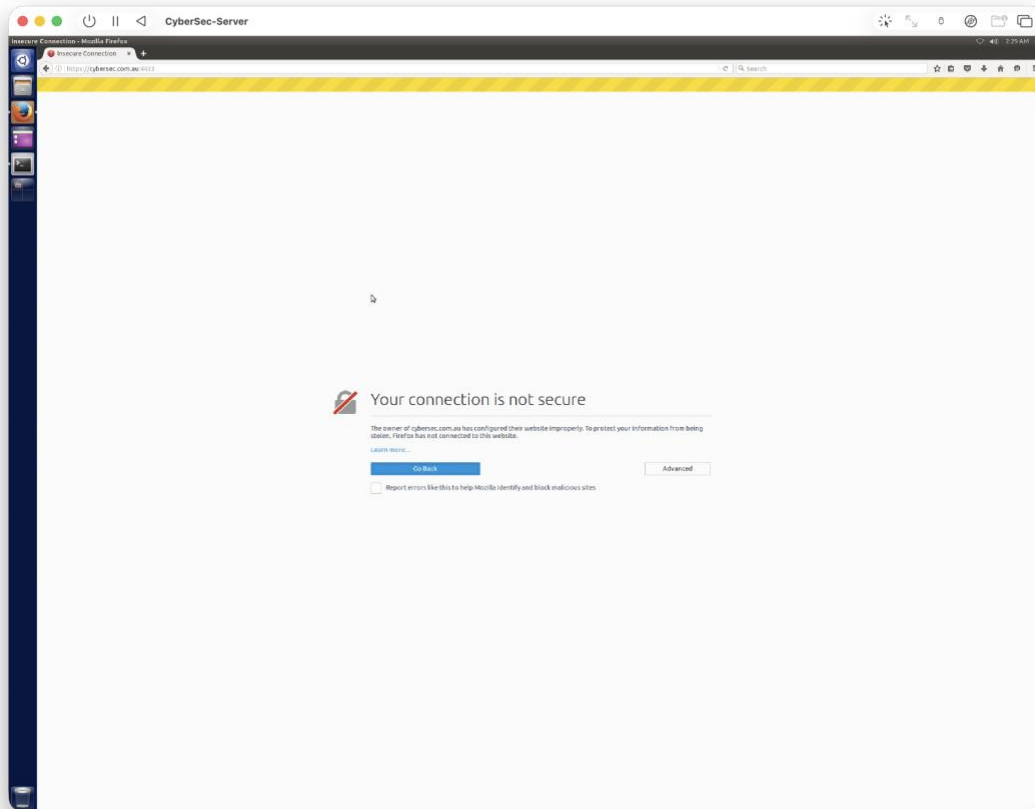


Fig 13. "Your connection is not secure - the owner of cybersec.com.au has configured their website improperly". This is the browser correctly refuse to trust an unknown CA.

Importing and trusting the CA

So i imported ca.crt into Firefox (Preferences → Advanced → Certificates → View Certificates → Authorities → Import) and tick "Trust this CA to identify websites".

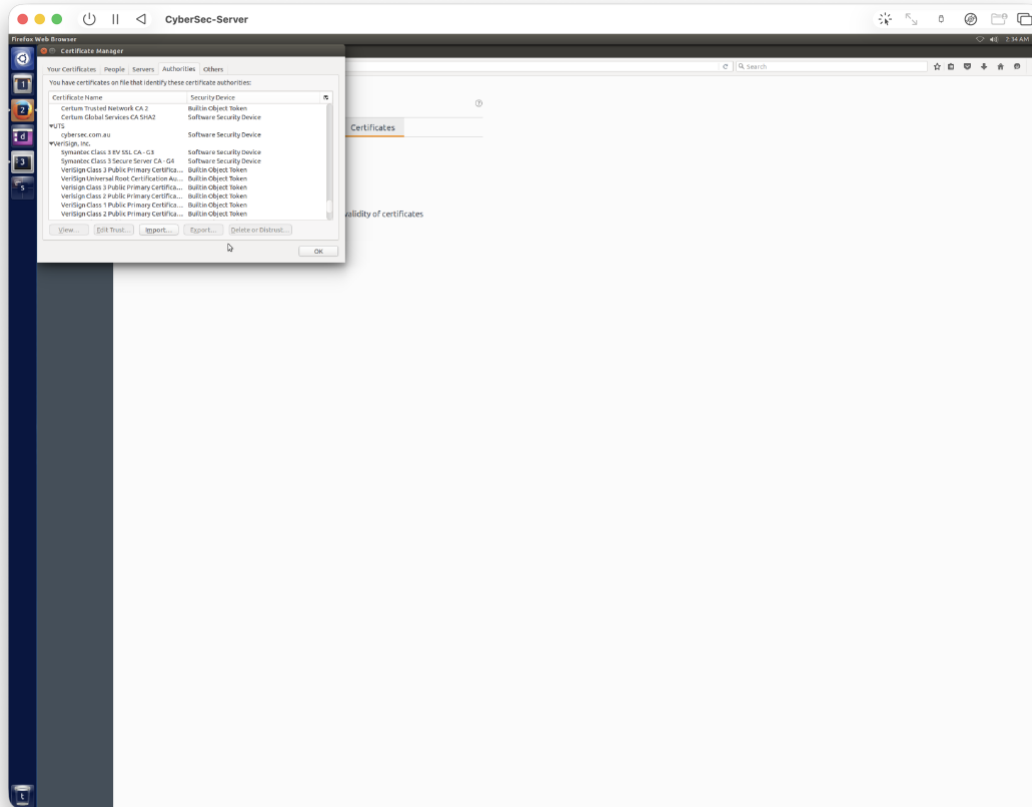


Fig 14. The Certificate Manager (Authorities) is now showing **UTS** → **cybersec.com.au**, so my CA is trusted.

After trusting the CA

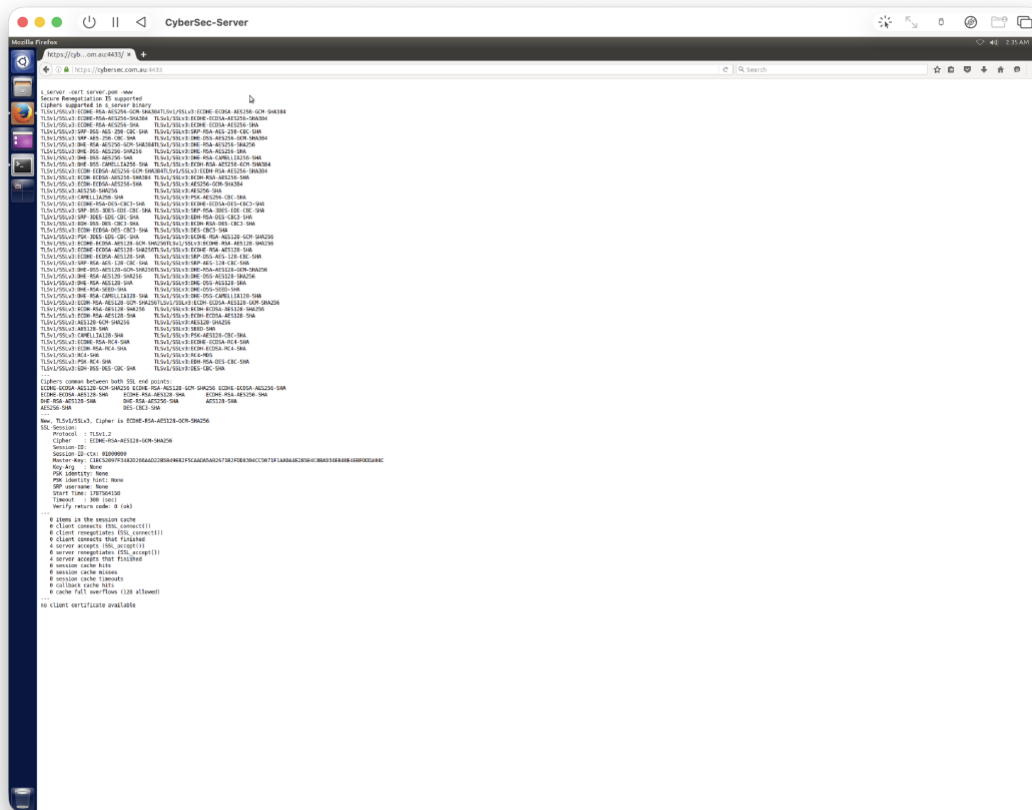


Fig 15. Reloading <https://cybersec.com.au:4433> now load with a padlock and a valid TLS session (TLSv1.2, ECDHE-RSA-AES128-GCM-SHA256, "Verify return code: 0 (ok)").

Observation: before i import my CA, Firefox block the site with the untrusted certificate warning; after i add ca.crt as a trusted authority, the exact same site load with a padlock and no warning at all. This show the core idea of the web trust model - a browser will trust any certificate that is signed by a CA it already trust. But it also show the danger: the Lenovo Superfish (2015) case is exactly this checkbox but abused in a big scale, a self-signed root CA with a recoverable private key was pre-installed on the laptops so the adware could silently intercept all the HTTPS traffic without any warning showing to the user.